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A Large Language Model based Multi-agent Framework for Goal Oriented Controller Design in Power Electronics

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***Abstract*—Power electronics, a critical component in modern power systems, face several challenges in control design, including model uncertainties, and lengthy and costly design cycles. This paper is aiming to propose a Large Language Models (LLMs) based multi-agent framework for objective-oriented control design in power electronics. The framework leverages the reasoning capabilities of LLMs and a multi-agent workflow to develop an efficient and autonomous controller design process. The LLM agent is able to understand and respond to high-level instructions in natural language, adapting its behavior based on the task's specific requirements and constraints from a practical implementation point of view. This novel and efficient approach promises a more flexible and adaptable controller design process in power electronics that will largely facilitate the practitioners.**

Keywords—objective oriented control design, power electron- ics, microgrids, large language models

I. INTRODUCTION

In modern power systems, power electronic devices usually serve as interfaces between renewable energy resources and loads [1]. Such devices play a pivotal role in regulating energy conversion and distribution [2]. Normally, it is fundamentally to design controllers based on the parameters of power electronic equipment and load fluctuations. However, privacy issues and aging of environmental impacts cause obtaining accurate parameters of power electronic components particularly difficult. These challenges may complicate the traditional controller design process, whether based on large-signal or small-signal analysis [3].

Traditional controller design process in power electronics generally rely on pre-established mathematical models of the system. These methods usually employ an iterative trial- and-error process, involving extensive modeling, tuning, and testing. Although these methods are widely used in practice, inevitable model variations, uncertainties, and unmodeled dy- namics may lead to inefficiencies in the design process [4].

Recently, model-free control methods have become a trend for power electronics as reported in the literature, which bypass the demand for explicit modeling and tuning, and virtual impedance control schemes [5]. While model-free control methods in power electronics have shown promises in addressing the challenges of traditional controller design, however, they are also facing some critical issues [6]. One of the main challenges is the implementation of the controller and safety constraints. Model-free control methods are designed without the explicit modeling and tuning process, focusing instead on achieving specific performance objectives However, the implementation of these objectives, such as reference voltage regulation in power electronics, still requires human intervention. In light of these challenges, there is an imperative need for further research and development in high-level control for power electronic systems, aiming to improve in areas such as objective definition, data handling, and system testing.

The objective-oriented control system (OOCS) is a novel concept that has been proposed to address the challenges associated with traditional control system design. The fundamental idea behind OOCS is to shift the design paradigm from "How" to "What" [7]. This means that instead of instructing the control system on how to achieve a specific objective, the designer specifies what the objective is, and the control system figures out how to achieve it. This can significantly reduce the time and resources required for control design. Therefore, the objective-Oriented Control System can simplifies the design process, enhances adaptability, and can potentially improve system performance.

In this study, a Large Language Models (LLMs) based agent framework is proposed for the objective-oriented control design in the field of power electronics. The framework leverages the reasoning capabilities of the LLMs agent and a multi-agent

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workflow to develop an efficient power electronics controller design framework. This makes the LLMs-based Agent Framework a promising tool for

II. Objective Oriented Controller Design

A. Multi-Agent Framework Architecture

In the design of the objective oriented Controller, we propose a multi-agent collaboration framework to decompose the objective Oriented controller design process into multiple sub-tasks, thereby achieving automation from task planning, modeling, controller design, simulation, to evaluation and verification.

The basic process for objective-oriented controller design is as follows and the workflow is shown as Figure 1:

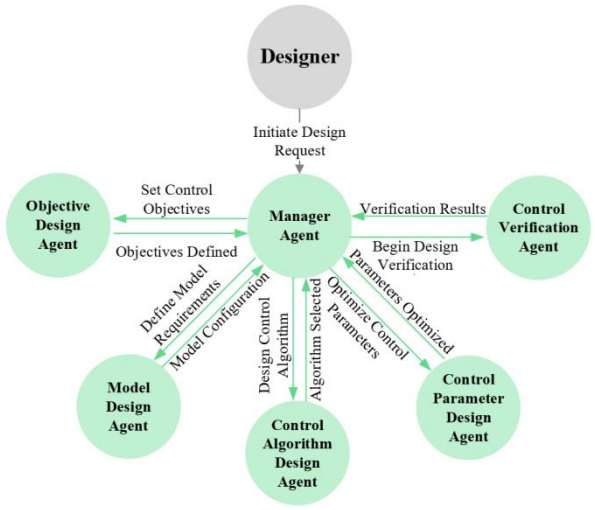


Fig. 1. Workflow of Objective Oriented Controller Design

The objective designer translates the task into control objectives, targets, and performance constraints. The manager, as the system’ s core, invokes various agents (e.g., control objectives, algorithms) to analyze these requirements and obtains relevant data. With the gathered information, the manager simulates the controller design (including selection and parameters). The evaluator then reviews the results, providing feedback or suggesting modifications to meet the task requirements.

B. Agent Role Design

The roles of these agents are designed as follows and the flow char of objective oriented controller design is shown in Figure 2.

The Manager Agent’s primary role is to orchestrate the objective-oriented controller design workflow, which includes identifying relevant questions, forwarding these inquiries to the appropriate agent, and summarizing the results. The Objective Design Agent needs to set the control objectives of the system according to its actual needs and corresponding performance indicators. The Model Design Agent is responsible for establishing a simulation model of the control object, which helps understand the system’s dynamic behavior and provides necessary information for the controller’s design. The Control Algorithm Design Agent selects the appropriate control algorithm based on the control objectives and performance indicators. The Control Parameter Design Agent is responsible for designing the controller’s parameters, such as the proportion, integral, and derivative coefficients for a PID controller. The Control Verification Agent verifies the performance of the controller through experiments, adjusting the parameters or changing the control strategy based on the experimental results if the performance does not meet expectations.

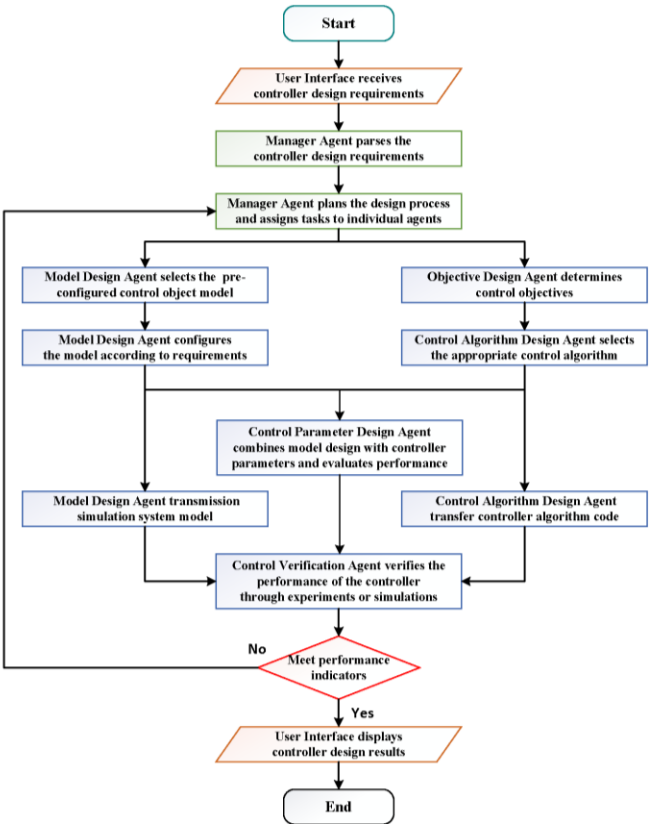


Fig. 2. Flow chart of Objective Oriented Controller Design.

III. Initial Implementation

Figure 3 showcases the initial implementation of our proposed multi-agent collaboration framework for task-oriented controller design. The detailed implementation of the core agent is described as follows.

A. Model Design Agent

The primary task of the Model Design Agent is to select simulation models and generate the Parameter for power electronic devices. It receives parsed design requirements from the Manager Agent. The agent then processes these

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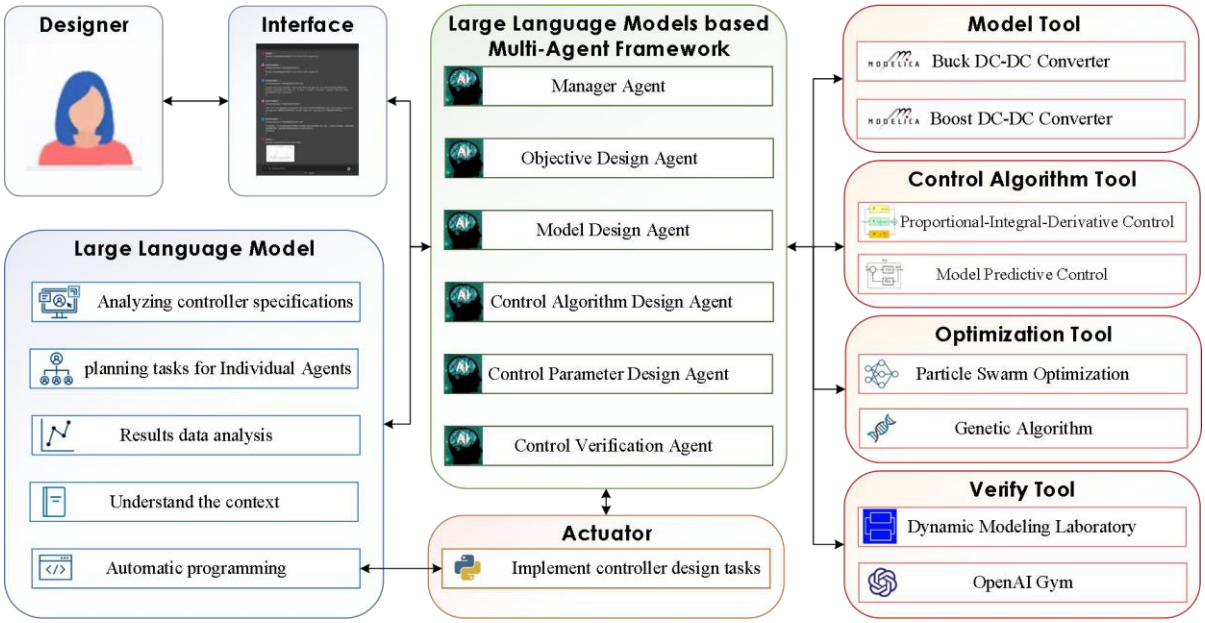


Fig. 3. Initial implementation of the multi-agent framework

requireents and selects the most appropriate Modelica model template using the Modelica model tool.

1) Tool: The Modelica model tool is a key tool in this process [8]. It may cover different types of converters, including buck, boost, and buck-boost converters. These templates serve as the starting point for model design, providing parameters that can be adjusted according to specific design requirements.

2) Output: This Modelica model file address is passed on to the Simulation Agent. It serves as the foundation for subsequent simulation verification and parameter optimization.

B. Controller Design Agent

The primary task of the Controller Algorithm Agent is to generate controller algorithm code automatically using the Controller Algorithm Tool.

1) Tool: The Controller Algorithm Tool empowers the Controller Algorithm Agent to generate preliminary controller algorithm code based on the design requirements, such as PID controller code. This tool references controller algorithm design templates to ensure the generated code is robust and efficient.

2) Output: The Controller Algorithm Agent hands over the controller algorithm code to the Controller Simulation Agent to ensure the controller meets the performance.

C. Controller Parameter Design Agent

The Controller Parameter Design Agent employs intelligent optimization algorithms to optimize controller parameters, where the optimization objectives and constraints are based on parameter design provided by the Target Design Agent, guiding this optimization process.

1) Tool: In terms of tool usage, the Controller Parameter Design Agent selects the optimization algorithm (such as Genetic Algorithm and Particle Swarm Optimization (PSO) algorithm [9] through the Optimization Tool and generates the final optimized parameter function.

2) Output: The agent passes the optimized parameter function class code to the Simulation Verification Agent.

D. Controller Verification Agent

The primary task of the Controller Verification Agent is to combine model files, control algorithms, and control parameter optimization functions to perform model simulation verification through the Open AI Gym toolbox [10].

1) Tool: The Gym toolbox is a potent simulation tool that allows the Controller Verification Agent to simulate controller performance under actual operating conditions. During this process, the agent collects performance data reflecting the controller's performance under actual operating conditions.

2) Output: Upon completion of the simulation, the Controller Verification Agent outputs performance indicators such as steady-state error, overshoot, and response time. An excellent style manual for science writers is [7].

These indicators are then passed on to the Objective Design Agent. If the simulation results do not meet the performance indicators required by Objective Design Agent, the unmet indicators are returned to the Manager Agent for task reallocation. However, if the performance indicators meet the requirements of the Objective Design Agent, the simulation verification data is passed on to the Manager Agent.

IV. Demonstration On DC-DC Boost Converter

To demonstrate the effectiveness of the proposed framework, a case study of task orient controller design is conducted for DC-DC Boost Converter.

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A. Controller Design Promp

In the process of designing objective oriented controller prompts, we need to specify the type of power electronic device for which the controller is being designed, the key parameters of the device, and the objectives that need to be achieved by the controller design. The design prompts for a DC-DC boost controller is shown in Figure 4.

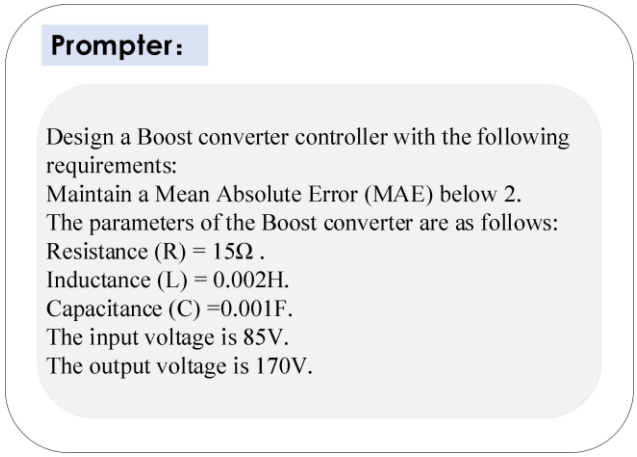


Fig. 4. Designing objective oriented controller prompts.

B. Tools Setup

In this experiment, we have implemented several tools to facilitate the design and optimization of our DC-DC Boost controller. These tools are designed to help us implement the controller algorithm, optimize the controller, and design the model of the DC-DC Boost circuit.

1) Controller Algorithm Tool: We have implemented a Proportional-Integral-Derivative (PID) control algorithm in our controller algorithm tool.

2) Optimization Tool: Our optimization tool employs the PSO algorithm.

3) Model Tool: For the model design, we have implemented a DC-DC Boost circuit model. This model helps us to simulate the behavior of the DC-DC Boost converter and test our controller under different conditions.

C. Results

As shown in Figure 5, the results of our experiment demonstrate the effectiveness of our multi-agent Framework approach for task-oriented controller design of a DC-DC converter.

V. Conclusion

In this work, we propose a novel Large Language Model (LLM)-based multi-agent collaboration framework for objective-oriented controller design in power electronics. Unlike traditional controller design that focuses on meeting design indicators, our approach utilizes multi-agent collabo- ration to realize the target tasks of controller design, thereby achieving automated design of power electronic controllers. Finally, we applied it to the objective-oriented design of a DC- DC boost converter controller, which verified the feasibility of our framework. This new approach opens a promising direction for the controller design and optimization of power electronics. It will significantly contribute to the advancement of controller design in the field of power electronics.

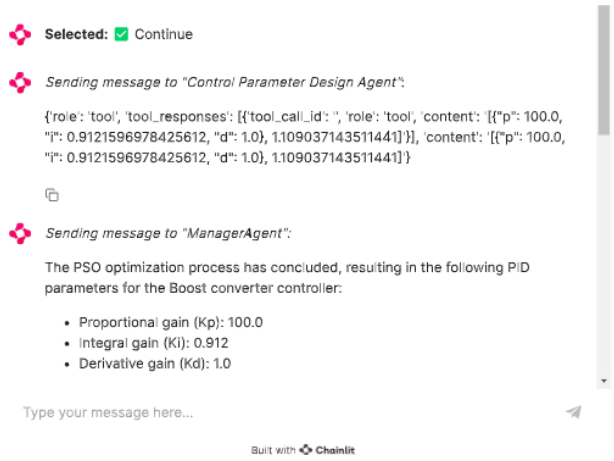


Fig. 5. Results of DC-DC boost converter controller Design

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